

Thinking Note - ThesisTrail

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Product: ThesisTrail - auditable NIFTY research workflow

1. Starting interpretation

The seed question was: **"Does buying NIFTY after a sharp fall work?"**

It sounds simple, but almost every word hides a research choice:

Phrase	Why it is ambiguous
NIFTY	Index research series vs ETF vs futures; only a proxy is tradable
buying	Timing (open/close), direction, and size are all unspecified
sharp fall	Absolute %, percentile, or volatility-scaled; lookback and session definition
work	Absolute profit, win rate, vs baseline, risk-adjusted, or economic significance

I treated the sentence as a prompt for clarification, not as a finished experiment.

2. User statement / uncertainty / decision table

Topic	Decision locked for this prototype
Instrument	NIFTY 50 index as research series; ETF/futures stay visible but unsupported
Sharp fall	Daily close-to-close return $\leq -2.0\%$ (-0.02), observed after the close
Entry	Next trading session open (not same-close)
Exit	Close of the fifth session counting entry as session 1 ($\text{exitIndex} = \text{entryIndex} + 4$)
Cost	10 bps illustrative round-trip ($\text{costRate} = \text{bps} / 10_000$)
Primary outcome	Median event net - median unconditional five-session baseline net
Event overlap	Ignore new signals while an event trade is open; count exclusions
Baseline overlap	Baseline windows may overlap (descriptive reference, not a portfolio)

3. Why these defaults

I chose **transparent, pre-registered** defaults rather than historically optimized ones. Next-open entry avoids treating a completed close signal as if same-close execution were available. A five-session inclusive hold is short enough to inspect and easy to audit. Medians reduce the pull of extreme events relative to means. The unconditional baseline answers "better than ordinary five-session holdings?" rather than "positive vs zero?" Costs stay illustrative because no tradable proxy was specified.

4. Data correction

I intended the inclusion window to start **2007-01-01**. Verified Kaggle coverage inside that window begins **2007-09-17** (source jumps from 2006-12-29). I corrected the **effective** start to 2007-09-17 **before any backtest result existed**, did not fabricate missing sessions, and treated this as data-integrity work-not outcome tuning. Effective period: **2007-09-17 through 2025-12-31**.

5. Risks and mitigations

Risk	Mitigation in the product
Ambiguity	Four clarification groups + overall confirmation before run
Look-ahead	Signal at close; entry next open
Instrument mismatch	Explicit index / non-tradable caveat; unsupported proxies selectable but blocking
Costs / slippage	Illustrative bps only; never claimed as verified broker cost
Overlap	Event suppression with exclusion counts; baseline may overlap by design
Overfitting	No parameter search; sensitivity reserved as future work
Provenance	Trace states: user stated -> proposed -> confirmed -> derived
Weak evidence	Locked classification strings; no p-values invented
Regime change	Single full-sample descriptive study; next tests propose splits
Causal claims	LEARN separates evidence, interpretation, and non-claims

6. Result and interpretation

Under the confirmed assumptions the locked engine reports **128** executed events (from **194** qualifying signals; **66** overlap exclusions) against **4,483** baseline windows on the fixed 4,487-session snapshot.

- Event median net ~ **-0.06%** (-0.0006313155503738779 as a decimal ratio)
- Baseline median net ~ **+0.15%** (0.0014686655804537771)
- Primary delta ~ **-0.21 percentage points** (-0.002099981130827655)
- Classification: **not_supported**

The observed sample does **not** support the sharp-fall hypothesis under these assumptions. I am not claiming the opposite strategy works, that the result generalizes to every regime, or that the finding is statistically significant in a formal hypothesis-test sense. It is a descriptive comparison under explicit assumptions.

7. Deliberately rejected scope

Auth/database, live broker or market APIs, same-close entry, parameter optimization for nicer history, multi-agent orchestration, AI-generated metrics or conclusions, and a large generic dashboard were rejected so the assessment stays about research honesty, not feature sprawl. Optional AI interpretation stays outside the evidence path.

8. What to test next

A tradable ETF or futures proxy with realistic costs; nearby-parameter sensitivity without optimization; regime separation (for example crisis vs calm years); out-of-sample splits; and instrument-specific friction studies that keep the same provenance discipline.